

1. First Non-Repeating Character • Find the first non-repeating character in a string using HashMap.

Code:

```
import java.util.*;

public class FirstNonRepeating {
    public static Character findFirstNonRepeating(String str) {
        Map<Character, Integer> map = new HashMap<>();

        for (char ch : str.toCharArray()) {
            map.put(ch, map.getOrDefault(ch, 0) + 1);
        }

        for (char ch : str.toCharArray()) {
            if (map.get(ch) == 1) {
                return ch;
            }
        }

        return null;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the string: ");
        String input = sc.next();
        System.out.print("The first non repeating character: ");
        System.out.println(findFirstNonRepeating(input));
    }
}
```

Output:

```
<terminated> Demo [Java Application] C:\Users\Le
Enter the string: racecar
The first non repeating character: e
```

2. Remove duplicate characters from string.

Code:

```
import java.util.*;

public class RemoveDuplicates {
```

```

public static void main(String[] args) {

    Scanner scanner = new Scanner(System.in);

    System.out.print("Enter a string: ");

    String input = scanner.nextLine();

    HashSet<Character> set = new HashSet<>();

    for (char c : input.toCharArray()) {

        set.add(c);

    }

    StringBuilder result = new StringBuilder();

    for (Character ch : set) {

        result.append(ch);

    }

    System.out.println("Result: " + result.toString());

    scanner.close();

}
}

```

Output:

```

<terminated> RemoveDuplicates
Enter a string: racecar
Result: race

```

3. Find element with maximum frequency.

Code:

```

import java.util.*;

```

```
public class MaxFrequency {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter a string: ");

        String input = scanner.nextLine();

        if (input.isEmpty()) {

            System.out.println("String is empty.");

            return;

        }

        HashMap<Character, Integer> counts = new HashMap<>();

        for (char c : input.toCharArray()) {

            counts.put(c, counts.getOrDefault(c, 0) + 1);

        }

        char maxChar = ' ';

        int maxCount = -1;

        for (Map.Entry<Character, Integer> entry : counts.entrySet()) {

            if (entry.getValue() > maxCount) {

                maxCount = entry.getValue();

                maxChar = entry.getKey();

            }

        }

    }

}
```

```
System.out.println("The element with maximum frequency is " + maxChar + " appearing " + maxCount + " times.");
```

```
scanner.close();
```

```
}
```

```
}
```

Output:

```
<terminated> MaxFrequency [Java Application] C:\Users\Lenovo\p2\pool\plugins\c
Enter a string: Application
The element with maximum frequency is 'p' appearing 2 times.
```

4. Sort a map by keys

Code:

```
import java.util.*;
```

```
public class SortMapByKeys {
```

```
    public static void main(String[] args) {
```

```
        Map<String, Integer> unsortedMap = new HashMap<>();
```

```
        unsortedMap.put("Zebra", 10);
```

```
        unsortedMap.put("Apple", 50);
```

```
        unsortedMap.put("Mango", 30);
```

```
        unsortedMap.put("Banana", 20);
```

```
        TreeMap<String, Integer> sortedMap = new TreeMap<>(unsortedMap);
```

```
        System.out.println("Sorted Map by Keys:");
```

```
        for (Map.Entry<String, Integer> entry : sortedMap.entrySet()) {
```

```
            System.out.println(entry.getKey() + " : " + entry.getValue());
```

```
        }
```

```
}
```

```
}
```

Output:

```
<terminated> SortMapByKey [Java /  
Sorted Map by Keys:  
Apple : 50  
Banana : 20  
Mango : 30  
Zebra : 10
```